GLOBALMART SALES FINANCIAL ANALYSIS

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Business Understanding

Business Overview

GlobalMart is a multinational retail company offering a diversified range of products such as furniture and technology. Its operations span in multiple states of the U.S.A including California and Florida. The company's sales data given provides valuable insights into transactional timelines and financial metrics, including revenue, quantity sold, discounts and profit. When a customer places an order, GlobalMart's system automatically logs this transaction and generates a unique identifies, Order ID. This serves as a primary reference for tracking each individual order. The system records the order date and captures relevant details, including the products purchased, state being delivered, quantities and applied discounts. This is what makes up the sales data provided.

Problem Statement

As a newly onboarded data analyst for GlobalMart, I have been tasked to:

Explore, analyse and visualise the company's sales data.

My analysis will support key decision-makers in ultimately guiding strategic planning and budgeting through data supported findings I intend to derive.

Main Objective

My main objective will be to identify growth opportunities, uncovering trends and addressing challenges in sales performance.

Specific Objectives

Along with my main objectives, I also intend to;

i. Cleaning, transforming and preparing data for analysis in Power BI.
ii. Calculating critical metrics such as Total Sales, Total Profit and Profit Margins.
iii. Creating interactive and visually appealing dashboards that present key insights.
iv. Identifying actionable recommendation to enhance sales performance and profitability.

Data Understanding

Data Characteristics

The dataset given contains GlobalMart's sales information from 2022 to 2024 which includes:

i. Order ID: Unique identifier for each order
ii. Order Date: Date the order was placed
iii. Ship Date: Date the order was shipped
iv. Region: Geographical region (e.g., East, West, Central, South)
v. State: State within the region
vi. Category: Product category (e.g., Furniture, Office Supplies, Technology)
vii. Sub-Category: Sub-category of the product (e.g., Chairs, Binders, Phones)
viii. Product Name: Name of the product
ix. Sales: Revenue generated from the sale
x. Quantity: Number of units sold
xi. Discount: Discount applied (in percentage)
xii. Profit: Profit earned from the sale

Descriptive Statistical Analysis

I did a descriptive statical analysis on quantitative columns which includes Sales Revenue, Profit and Quantity as seen below.

Summary on Sales Revenue		*Summary on Profit*		*Summary on Quantity*	
Mean	2293.827003	Mean	339.930623	Mean	5.39465875
Standard Error	1510.729572	Standard Error	11.5737593	Standard Error	0.15391683
Median	492.78	Median	327.02	Median	5
Mode	#N/A	Mode	215.45	Mode	4
Standard Deviation	27733.30838	Standard Deviation	212.465978	Standard Deviation	2.82553745
Sample Variance	769136393.7	Sample Variance	45141.7918	Sample Variance	7.98366186
Kurtosis	311.7087326	Kurtosis	-0.5802106	Kurtosis	-1.1453216
Skewness	17.43521839	Skewness	0.33253118	Skewness	0.05206095
Range	499979.46	Range	971.08	Range	9
Minimum	20.54	Minimum	-25.55	Minimum	1
Maximum	500000	Maximum	945.53	Maximum	10
Sum	773019.7	Sum	114556.62	Sum	1818
Count	337	Count	337	Count	337
Confidence Level(95.0%)	2971.679673	Confidence Level(95.0%)	22.7661561	Confidence Level(95.0%)	0.30276201

Figure 1: Descriptive Statistics Summary on Sales Revenue, Profit and Quantity

Sales Revenue

i. Extreme Right Skew (Skewness = 17.44)
Sales are heavily influenced by a few large transactions. This indicates revenue concentration among a few key orders.

ii. High Variability (Standard Deviation ≈ $27,733)
The large spreads suggests inconsistent sales patterns. This can be better examined through checking the sales revenue by product, region or time to identify causes of volatility.

Profit

i. Presence of losses (Min = -$25.55)
A portion of transactions results in negative profit. I should check for over-discounting in order to see if there is a possible connection to this.

ii. Low Variability (Standard Deviation ≈ $212)
This indicates cost structures are more predictable than revenue streams.

Quantity

i. Consistent ordering behaviour (Mean ≈ Median ≈ Mode)
Most customers order between 4 – 6 items.

ii. Tight Range (1-10 units)
This implies that almost all order fall within a narrow window which may suggest that GlobalMart sells to individuals or small enterprises.

Data Preparation

Data Cleaning

Before analysis could be conducted on the dataset, I thoroughly reviewed it in Microsoft Excel before extracting and using the data in Power BI. The following steps were taken to prepare the data:

i. Duplicate Removal: The Order ID column serves as unique transaction identifier. Using Excel's Remove Duplicates function, I deleted 10 redundant entries to ensure each order is counted once.

ii. Standardised Column Names and Labels
   a) Renamed Ship Date to Shipping Date for clarity
   b) Renamed Sales to Sales Revenue for clarity
   c) Applied Title Case consistently across all column headers and data such as in the Region column using Excel's PROPER () function.

iii. Handling Missing Values: Rows with missing critical data points such as Order Date, Region or Sales Revenue were removed. A large number of rows were deleted, which may slightly affect the completeness.

iv. Data type Standardisation: All columns were put under consistent formatting as follows:
   a) Text: Order ID, Region, State, Category, Sub-Category, Product Name
   b) Date: Order Date, Shipping Date
   c) Currency (Dollars and 2 decimal points): Sales Revenue, Profit
   d) Whole Number: Quantity
   e) Percentage: Discount

Disclaimer

While cleaning, some rows were deleted due to missing data. This may introduce marginal variance in results.

Key Performance Indicators (KPIs)

To get better insights on the data given, I created several KPIs using DAX measures:

	KPI	Description
1	Total Revenue	Total sales revenue generated from all transactions
2	Total Profit	Sum of all profits after subtracting costs from revenue
3	Profit Margin	Highlights efficiency in converting revenue into profit
4	Total Quantities Sold	Aggregate count of all items sold
5	Sales by Region	Breakdown of revenue by state/region
6	Revenue YoY%	Year-over-year percentage growth of total revenue
7	Shipping Duration	Calculated as the number of days between Order Date and Shipping Date
8	Revenue PY	Total revenue earned in the same period during the previous year

Data Modelling

In order to analyse my data in Power BI, I first extracted my data and transformed it. This process is supposed to help me identify primary and foreign keys, creating normalised tables and establish a clear star schema to optimise my data.

Fact and Dimension Tables

The central Fact Table, Sales, stores transaction-level data which are quantifiable. From this table, I derived dimension tables which can relate through foreign keys. To ensure normalisation and reduce redundancy, I created the following Dimension Tables:

Product Dimension Table

- The table contains detailed product metadata.
- Primary Key: Product ID
- Columns:
    i. Category
    ii. Product ID
    iii. Product Name
    iv. Sub-Category

Region Dimension Table

- The table contains location-specific data at both regional and state levels.
- Primary Key: Region_State ID (a concatenation of Region and State to ensure uniqueness)
- Columns:
    i. Region_State ID
    ii. Region
    iii. State

Date Dimension Table

- The date table spans from 01/01/2022 to 31/12/2024, covering the full range of transactional data. This enables flexible time intelligence functions and ensures all orders are accurately captured.
- Columns include:
    i. Date
    ii. Day of the Week
    iii. Month
    iv. Month Number
    v. Week Number Year
- The Date Table is connected to the Sales table via:
    i. An active relationship with Order Date
    ii. An inactive relationship with Shipping Date, which can be activated using the USERELATIONSHIP function

Measures Tables

All calculated measures were centralised into one table for easy tables management. This improves model organisation and allows easier reference when building visuals and dashboards.

Included Measures:

i. Total Shipping Duration
ii. Profit Margin
iii. Total Revenue
iv. Revenue YoY%
v. Revenue PY

Data Relationships and Cardinality

To establish logical connections between tables:

i. Sales → Product: One-to-Many (Product ID as foreign key in Sales)
ii. Sales → Region: One-to-Many (Region_State ID as foreign key in Sales)
iii. Sales → Date: One-to-Many (via Order Date; inactive for Shipping Date)

These relationships collectively form a Star Schema enabling efficient querying as shown below;

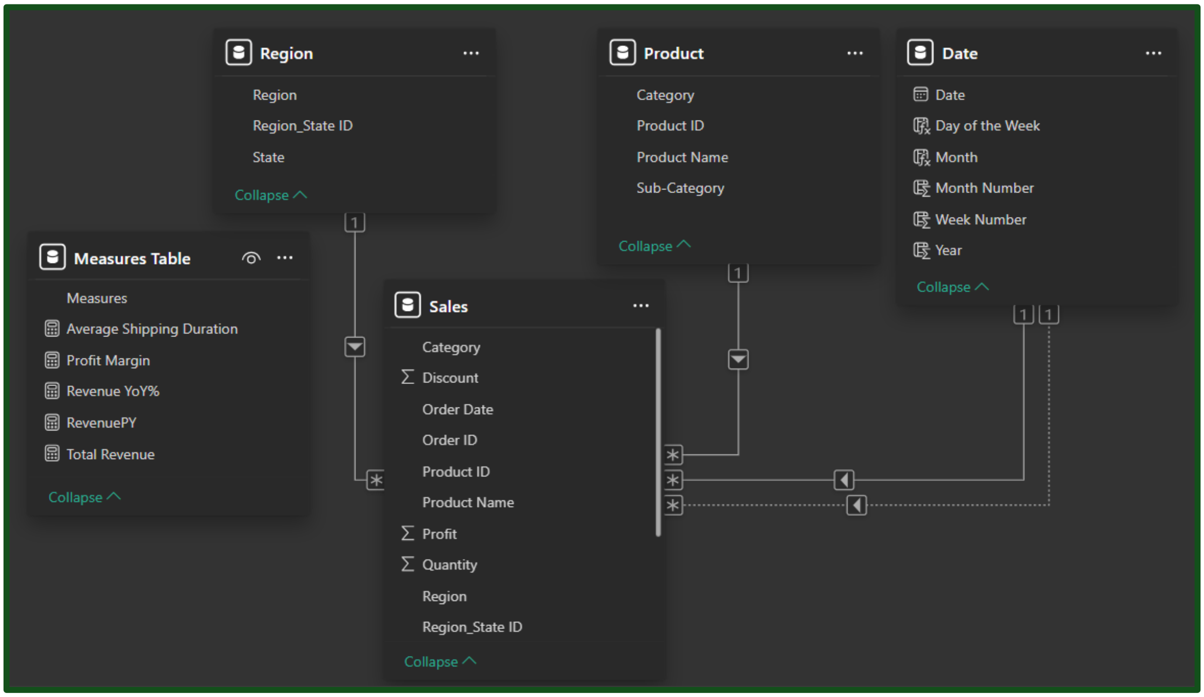


Figure 2: Star Schema and Relationship architecture

Cross-Filter Direction

All relationships are configured with single-directional filtering from dimension to fact tables. This minimises clashing of data ensuring performance optimisation.

Granularity

The dataset operates at a low granularity level. This enables high-level aggregated insights such as monthly sales or regional performance while avoiding unnecessary data complexity.

Report Modelling

Executive Summary Report

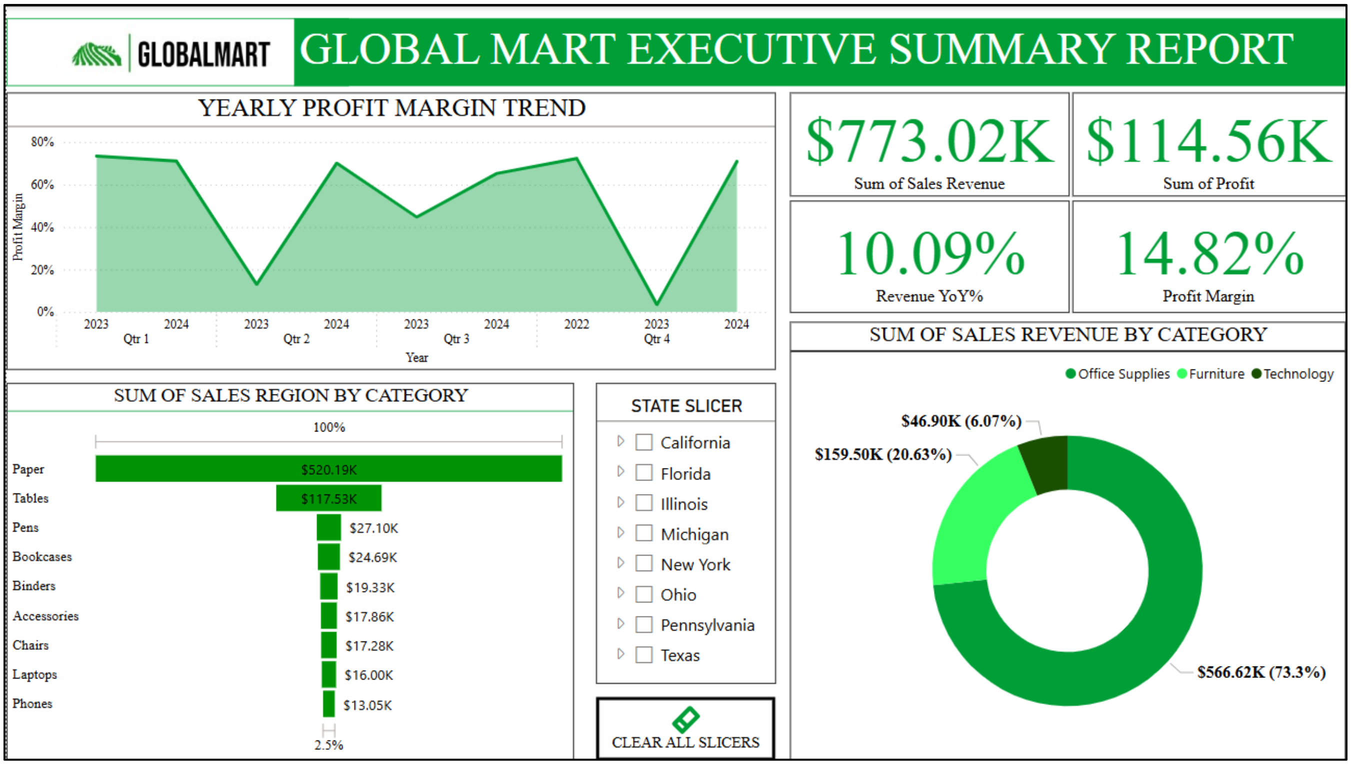
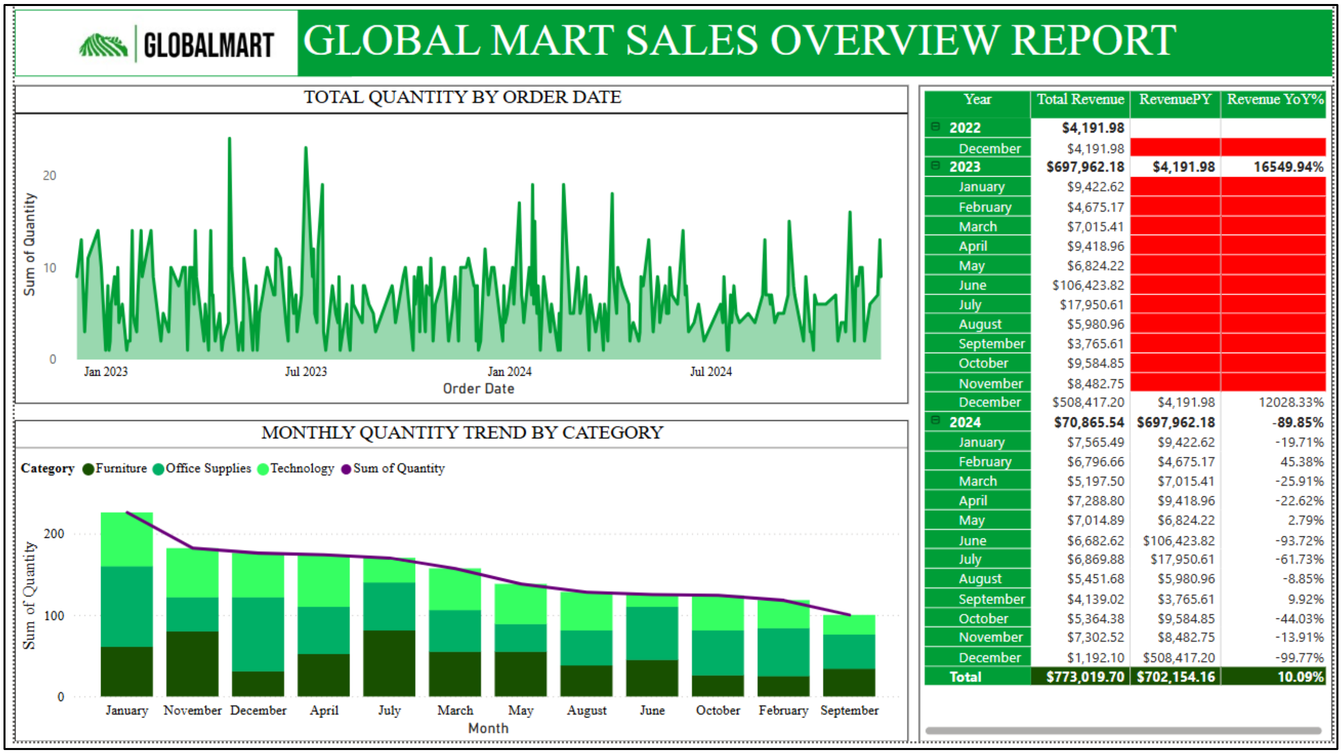


Figure 3: Executive Summary Report

Key Findings

- Office Supplies dominates, accounting for more than 70% of sales revenue. This indicates a heavy reliance on products such as paper, tables and pens.
- Technology is underperforming having the smallest share (6.07%) of the sales revenue.
- Profit Margin fluctuates between 20% and 80%, indicating high volatility.
- Profit margin (14.82%) is relatively modest considering the strong revenue, however, this highlights potential inefficiencies exist in cost management.
- Positive Year-Over-Year (10.09%) is encouraging but should be improved.

Sales Overview Report



Year	Total Revenue	RevenuePY	Revenue YoY%
2022	$4,191.98		
December	$4,191.98		
2023	$697,962.18	$4,191.98	16549.94%
January	$9,422.62		
February	$4,675.17		
March	$7,015.41		
April	$9,418.96		
May	$6,824.22		
June	$106,423.82		
July	$17,950.61		
August	$5,980.96		
September	$3,765.61		
October	$9,584.85		
November	$8,482.75		
December	$508,417.20	$4,191.98	12028.33%
2024	$70,865.54	$697,962.18	-89.85%
January	$7,565.49	$9,422.62	-19.71%
February	$6,796.66	$4,675.17	45.38%
March	$5,197.50	$7,015.41	-25.91%
April	$7,288.80	$9,418.96	-22.62%
May	$7,014.89	$6,824.22	2.79%
June	$6,682.62	$106,423.82	-93.72%
July	$6,869.88	$17,950.61	-61.73%
August	$5,451.68	$5,980.96	-8.85%
September	$4,139.02	$3,765.61	9.92%
October	$5,364.38	$9,584.85	-44.03%
November	$7,302.52	$8,482.75	-13.91%
December	$1,192.10	$508,417.20	-99.77%
Total	$773,019.70	$702,154.16	10.09%

Figure 4: Sales Overview Report

Key Findings

- Highly volatile monthly sales patterns. Alarmingly, across a span of three years, only 12 distinct order peaks are observed showing limited sales activity.
- Only three months, (December 2023, May 2024 and September 2024), register positive profit margins. Notably December 2023 recorded an anomalous spike with a profit margin of 12,028.33%. This artificially inflates the overall annual profitability as without this outlier, the business would have closed the year with a net loss.
- Order volumes peak in December and January, suggesting seasonal demand influenced by external factors such as holiday period or increased consumer spending.

Sales Detailed Report

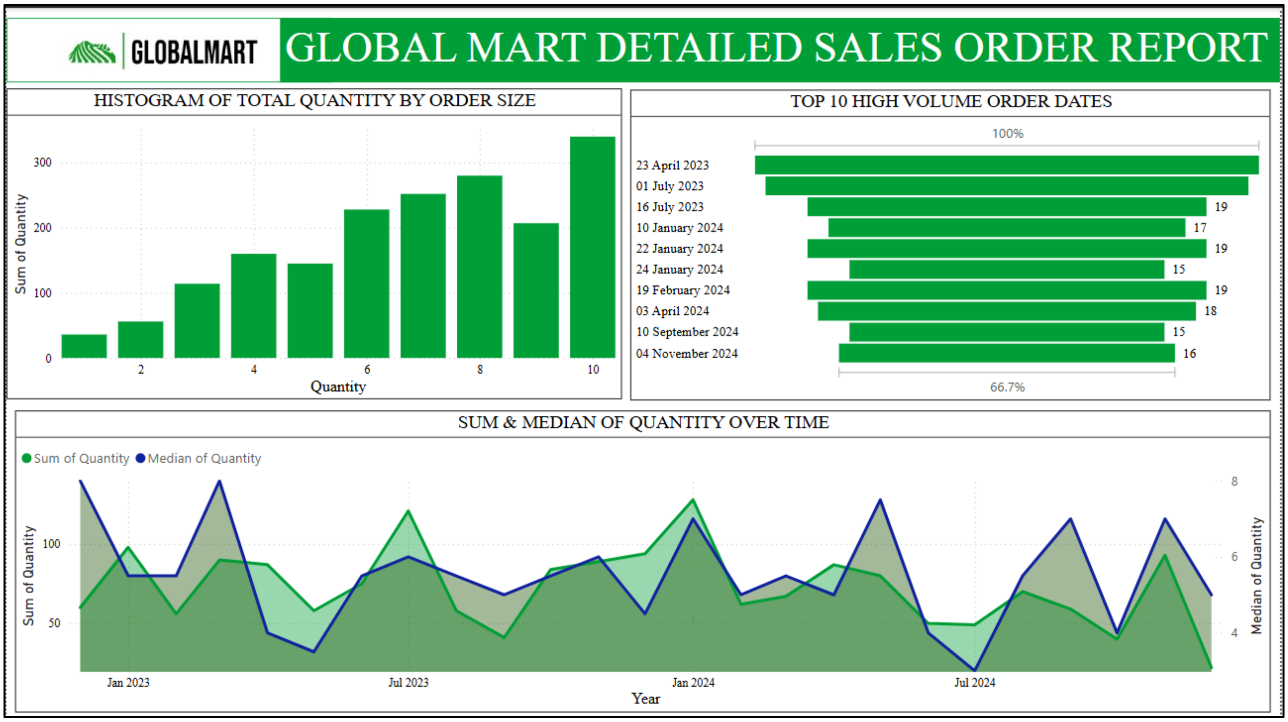


Figure 5: Detailed Sales Order Report

Key Findings

- Customer order behaviour is skewed toward small batches. This suggests B2C (Business to Customer than B2B (Business to Business).
- High volume spikes suggest occasional bulk buyers which are likely contract deals.
- Stable median values imply that majority orders remain predictable.
- Sales peaked on 23rd April 22023, 1st July 2023, 10th , 22nd and 24th January 2024.

Region Analysis Report

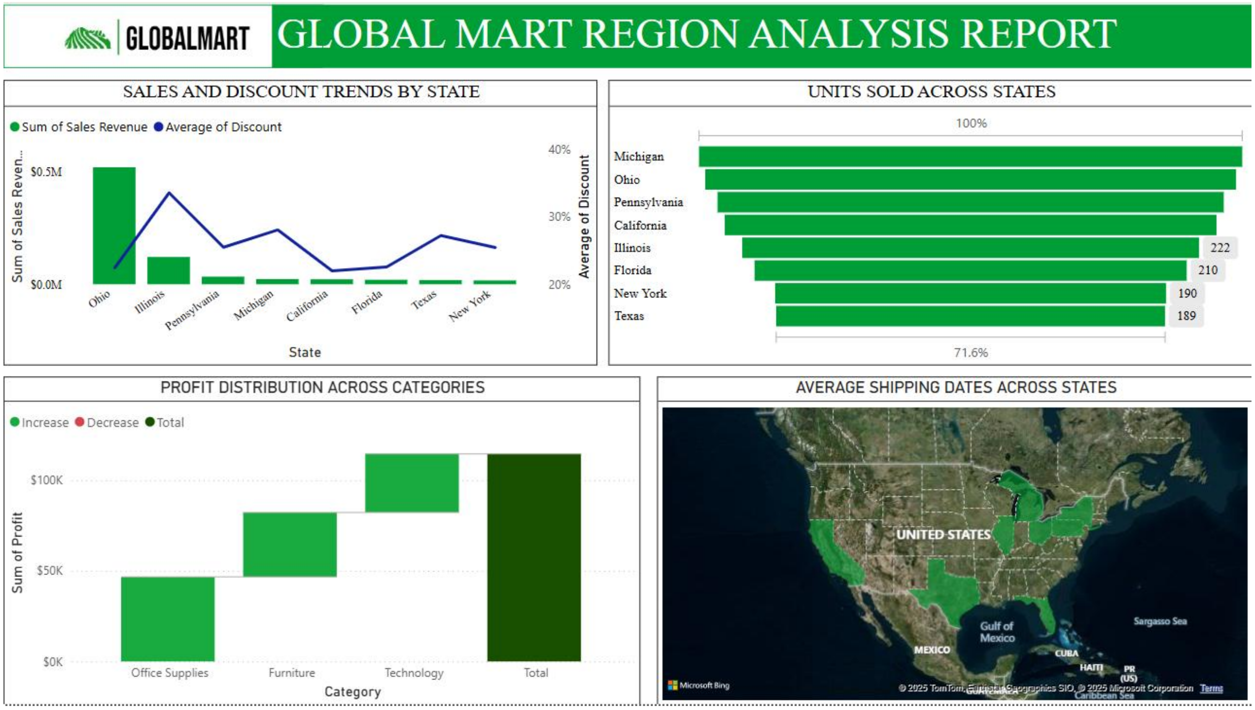


Figure 6: Region Analysis Report

Key Findings

- Ohio, Illinois and Pennsylvania lead in revenue.
- Ohio having a significantly huge lead than the rest and surprisingly the lowest average discount percentage.
- Office supplies has the greatest profit distribution in all regions followed by Furniture and technology.
- Michigan has the most units sold yet overtaken by Ohio by a huge margin in Sales revenue.
- All states have an average of 4 to 5 days in shipping dates which is relatively good

Evaluation

Main Objective Evaluation:

"Identify growth opportunities, uncover trends and address challenges in sales performance."

Achievement:

I. Numerous growth opportunities have been identified such as capitalising on strong state performance such as Ohio and optimising office supplies dominance.
II. Trends were uncovered in sales such as a volatile monthly sales pattern and dominant product category.
III. Challenges like low profit margins, tech underperformance and poor demand predictability are identified and analysed.

Specific Objectives

I. "Cleaning, transforming and preparing data in Power BI"

- Reports and insights have been derived from the 4 reports I have made.

II. "Calculating critical metrics such as Total Sales, Total Profit and Profit Margins"

- These Calculations have been made under the Measures table.
- Additional metrics such as Year-Over-Year growth were made.

III. "Creating interactive and visually appealing dashboards"

- This was not fully achieved as the creation of a dashboard needs Power BI Service which I do not have access to. In replace of this I made dashboard looking reports in Power BI which are well-structured and insightful.

IV. "Identifying actionable recommendations to enhance sales performance and profitability"

- I have gotten actionable insights such as addressing cost inefficiencies.

Recommendations

I. Ohio must be prioritised as the strategic growth hub

Ohio undeniably commands the highest revenue across all states with a huge margin while surprisingly maintaining one of the lowest average discount rate. This confirms superior pricing power and customer loyalty. The company must direct more inventory and marketing expenditure to this state to maximise Return On Investment (ROI).

II. Technology category must undergo immediate revitalisation

The technology segment contributes only 6.07% of the total revenue, despite being a high-value category. This level of underperformance cannot be allowed. The company must initiate a complete product audit and enhance product offerings to promote this category.

III. Michigan's Under-Monetisation Must be corrected

Although Michigan reports the highest volume of units sold as seen in Figure 6, it lags behind in revenue. This proves low average order value or excessive discounting. Sales in this state must be adjusted through upselling and adjusting discount strategies to fully monetise customer demand.

IV. Product Mix must be diversified to reduce over-reliance on office supplies.

With over 70% of revenue coming from office supplies, the company is dangerously dependent on a single category.

V. Seasonal Demand Spikes must be systematically capitalised on

The peak order periods, December and January, coincide with holiday seasons and increased disposable income. The company must institutionalise seasonal campaigns and dynamic pricing models to fully extract maximum value from these high-volume windows.